



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off - campus of Nitte (Deemed to be University)
First Semester M.Tech. (CBS) (CBCS) Degree Examinations
January 2023

22CBS101 – INTRODUCTION TO CYBER SECURITY AND SECURE CODING

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Briefly explain the characteristics of OSI reference model.	10	L*2	1	1
	b)	Explain the CIA framework of a cyber security model.	10	L2	1	1
2.	a)	Explain the working of a subnet masking with the help of an example.	10	L2	1	1
	b)	List and explain any five fundamental security design principles.	10	L2	1	1
		Unit – II				
3.	a)	List the various general objectives of software architecture and design.	10	L2	2	1
	b)	Describe Threat Analysis in detail.	10	L2	2	1
4.	a)	List the various security specific objectives of software architecture and design.	10	L2	2	1
	b)	Describe Architectural Vulnerability Assessment in detail.	10	L2	2	1
		Unit – III				
5.	a)	Briefly explain the following principles of software security in detail. i. Least privilege ii. Failing securely iii. Securing the weakest link iv. Defense in depth v. Separation of privilege	10	L2	3	1
	b)	Briefly explain the various attack patterns.	10	L2	3	1
6.	a)	Briefly explain the guideline for following the rules regarding Concurrency Management.	10	L2	3	1
	b)	List the advantages and disadvantages of various categories of software metric.	10	L2	3	1
		Unit – IV				
7.	a)	Briefly explain the various security coding practices to be used for input validation.	10	L3	4	2
	b)	Discuss database security in detail.	10	L2	4	2
8.	a)	Explain the best practices used for data protection.	10	L3	4	2
	b)	Briefly explain the various security coding practices to be used for authentication and password management.	10	L2	4	2
		Unit – V				
9.	a)	With an example explain the concepts of divisibility and prime numbers.	10	L2	5	1
	b)	Explain the arithmetic functions and congruence by giving example for each.	10	L2	5	1
10.	a)	Explain the following algebraic structures by giving an example for each: i. Groups ii. Rings	10	L2	5	1
	b)	Explain the quadratic residues and primitive roots in detail.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome